IF/THEN: rules that act

IoT · Level 3 · Lesson 2



Today we will

- Write automation rules as IF condition THEN action, with sensible thresholds.
- Use two thresholds so devices don't flicker, and make the system fail safe.
- Test rules against a day of data and judge comfort against energy.

Rules that act

- IF a condition, THEN an action
- Choose thresholds from logged data
- Two thresholds stop flicker
- When a sensor fails, fail safe

Rule cards through a school day

1. Warm-up (5 min): When should our fan run? Collect rules.
2. Teach (8 min): IF/THEN, thresholds and the flicker problem.
3. Play the day (12 min): Read out readings; devices stand when on.
4. Fix it (10 min): Two thresholds, and a plan for a silent sensor.
5. Wrap-up (5 min): Write one rule for a real school device.

Before class, write a day of readings on the board or a chart (time, temperature, light, tank level), with a stretch that wobbles around 30 °C and one hour when the tank sensor is silent. Pupils act the devices: no mains appliances are switched in class.

Let’s discuss

1. Which school device would you automate first, and why?
2. Who should choose the balance between comfort and electricity?
3. What should happen when a sensor goes silent at night?

Exit ticket

- Write a two-threshold rule for the tank pump, and say what it does if the sensor goes silent.